ARSENIC AUTOMATAION COMMUNITY FILTER



I built an IOT device with point to point communication, this involved mechanical and electronic parts.
The chemical storage tanks are manually filled with the chemicals. There are 2 water tanks with sensors to maintain the quantity of water. A PLC was used to set the automatic programming- where the predefined quantity of chemicals to automatically dispense, then automatic stirring would make a homogeneous mixture.

Process description
The automation process starts with receiving tap water and putting it inside the water tank. Then we have to dispense 3 chemicals in succession which are sodium bicarbonate potassium permanganate and ferric chloride to dispense all the 3 chemicals we used 3 peristaltic pumps which are10 MLS per minute 10 ml per minute and 55 ml per minute respectively. I built a contraption that would accurately dispense the 3 chemicals.
After dispensing each chemical we need to stir the mixture for that we made a pp rod with 6 hands, since PP doesn't react with these chemicals.
The stirrer will automatically move after the chemicals are dispensed and make it a homogeneous mixture.
To spin the contraption we used this makeshift motor which was derived from an industrial exhaust , the motor spins at 960 rpm but didn't provide enough torque so I also added a gear box with a worm gear that reduces the rpm to 50 and hence increases the torque.
After stirring we need to wait for a few minutes for the precipitates to form and for coagulation so that the water becomes arsenic free.
Water is then transferred from tank A to tank B, water is then ready to drink from tank B.

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Letter of Recommendation

For Mr. Aarav A. Jain

This is my pleasure to write about Mr. Aarav A. Jain who did internship with me during Summer 2025.

Mr. Aarav emailed me with a proposal to work with me a couple of months before meeting at Tezpur University during the summer. I was initially hesitant to allow him to work in my laboratory as he was too young to do research. However, I had to change my mind after he had met me in the university. Throughout my career, I have noticed, a student's dedication is more important than his talent for the kind of research I have been doing. I had been impressed by his enthusiasm and interest in reaching out to rural people through safe water which is my main research area.

During his internship Mr. Aarav contributed towards designing in developing a handheld bottle filter, developed by my research group, that removed arsenic other heavy metals and turbidity in addition to regulating the pH of water using the already patented Arsiron Nilogon method. This has enabled him to include his name among the inventors of a patent entitled, "Handheld portable filter for removal of arsenic, heavy metal ions and turbidity from drinking water", filed in the Indian Patent Office, with application number of 202531130896, dated 23 December 2025.

As I saw a proficiency in AI in Mr. Aarav, I advised him to make an automated dosing unit for a small community, say 200L-1000L Arsiron Nilogon filter. He was able to make it and successfully demonstrate it in a demo Arsiron Nilogon unit at the University campus. Mr. Aarav continues to be in touch with me after the internship and utilises some of his time in working towards helping rural people in getting arsenic-free safe water.

Finally, I can say that I have not only seen a dedication in Mr. Aarav to his work, but also has an inquisitive and talented mind which he proactively applies in whatever he does. We all need to nurture such a talented and dedicated boy.

09/4/2026

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